Amir Beck, Chapter 1 - Exercise 1.3

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1.3. Show that for any $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^{\mathbf{n}}$

$$||\mathbf{x} - \mathbf{z}|| \leq ||\mathbf{x} - \mathbf{y}|| + ||\mathbf{y} - \mathbf{z}||.$$

Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$, then

$$||\mathbf{x} - \mathbf{z}|| = ||(\mathbf{x} - \mathbf{y}) + (\mathbf{y} - \mathbf{z})|| \leq ||\mathbf{x} - \mathbf{y}|| + ||\mathbf{y} - \mathbf{z}||,$$

where in the last step we used the triangle inequality property of the norm.